Atta Ul Asad

M.S. Student in Artificial Intelligence, Lahore University of Management Sciences (LUMS)

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Research Interests

My research asks what aggregate accuracy hides: whether an AI second opinion checks a clinician's diagnosis or only echoes it, what compression removes, and whether post-training gains reach unseen problems.

Clinical LLMs and Human–AI Decision-Making. I study LLMs as clinical decision support, judged by their effect on the final decision, not standalone accuracy: when a second opinion adds independent judgment and when it echoes the clinician's working diagnosis. I measure harmful and helpful changes separately and study safeguards against anchoring, sycophancy, and automation bias that keep useful corrections.

Efficient Inference and KV-Cache Compression. I design training-free KV-cache eviction that keeps what future decoding needs, and audit what compression costs beyond accuracy, from faithfulness under KV-cache quantization to reasoning under low-bit weights, testing proposed fixes against simple baselines.

Post-Training for Self-Improving LLMs. I study RL with verifiable rewards (RLVR) and self-evolution: which training tasks, including self-generated ones, improve a model on problems it has not seen, through Challenger–Solver self-play, minimax-regret environment design, and influence-guided data selection.

Education

Lahore University of Management Sciences (LUMS) August 2026 – Present
Master of Science in Artificial Intelligence

- **Coursework** Foundations of AI, Mathematics for AI

COMSATS University Islamabad September 2022 – June 2026
Bachelor of Science in Computer Science

- **Dean's List of Honors** SGPA 3.98/4.00 in the third semester
- **Senior Design Thesis** AI Teaching Assistant System: A Multimodal Intelligent Assistant for Automated Student Support and Analytics
- **Relevant Coursework** Natural Language Processing, Machine Learning, Data Structures and Algorithms, Database Systems, Linear Algebra, Object-Oriented Programming

Research Experience

Lahore University of Management Sciences (LUMS), Independent Researcher August 2026 – Present
Project: Second Opinion or Echo? (first author; ACM FAccT 2027)

- Designed a controlled audit of LLM second opinions on 184 case reports whose working diagnosis was later overturned, built from 98,641 MultiCaRe narratives with hand-checked samples: each report is cut before the error and after later findings, and one varied sentence suggests a wrong diagnosis (documented, LLM-generated, or unrelated) or the final one.
- Found in five open-weight LLMs (4B–24B, three families) that the documented error turned 37–89% of correct answers wrong while the correct diagnosis fixed 39–92% of wrong ones, so agreement could not separate them; with later findings, the error still overturned 31–64% of correct answers in three models.
- Compared documented with LLM-generated errors: the latter swayed models at least as much, yet without any suggestion every model favored the documented error on the early record, a preference that audits built only on constructed errors would miss.
- Showed that an independence instruction, and thinking mode in Qwen3, weakened right and wrong suggestions alike: the help the instruction removed was 3.2–8.4× the harm it prevented, so with the two weighted equally it beat the plain request only if over 76–89% of suggestions were wrong.

Stanford AI Lab (SAIL), Stanford University, Research Collaborator January 2026 – Present
Collaborators: M. A. Mohsin and M. Umer

- **LORE-KV (ICLR 2027).** Co-authored a training-free KV-cache eviction method that samples short continuations from the frozen target model and scores each prompt token by its projected leave-one-out attention-output deletion cost, aggregated across sampled futures. At a 128-token budget it raised LongBench on Qwen2.5-14B from 45.49 to 48.24 and 16K RULER on Mistral-7B from 45.20 to 51.05; one sampled continuation recovers about 89% of the gain over the prompt-window control. Joint work with researchers at the University of Oklahoma, the University of Cambridge, the University of Houston, and Isara Labs.
- **Influence-Guided Self-Evolution and Game-Theoretic Environment Design for Language Models (in progress, targeting ICML 2027).** Post-training for self-evolving LLMs: Challenger–Solver self-play with minimax-regret environment design, plus influence-guided selection of RLVR data, to find which tasks, including self-generated ones, improve a model on unseen problems. With A. Bilal and M. A. Mohsin.

University of Oklahoma, Research Intern June 2026 – August 2026
Advisor: Dr. Dean F. Hougen | Project: Faithfulness Is Not Free (GroundLM @ EMNLP 2026)

- **KV-Cache Quantization in RAG.** Audited BF16, INT8, and INT4 quantization of offline document KV caches in RAG with Qwen2.5-7B-Instruct on RGB and HotpotQA, measuring accuracy (EM, F1) and faithfulness with HHEM, DeBERTa-v3 NLI, and an LLM judge.
- Showed that INT8 is near-lossless while INT4 reduces accuracy and, among answers that remain correct, over 90% of faithfulness changes are negative, a regression accuracy metrics miss; the harm grows under noisy retrieval and with more chunks (paired McNemar tests, OLS).

University of Malaya, Undergraduate Research Assistant July 2025 – September 2025
Advisor: Dr. Uzair Ishtiaq

- Built BioBERT-based clinical NLP pipelines for biomedical named-entity recognition, clinical text classification, and disease-related information extraction.

PUBLICATIONS

* denotes equal contribution. Author name in **bold**.

Under Review

[P1] Second Opinion or Echo? How Documented Misdiagnoses Steer Language Models ACM FAccT 2027
***A. U. Asad**, A. Bilal, R. M. S. Khan, D. F. Hougen* [PDF] [Code]

[P2] Monte Carlo Estimation for KV Cache Eviction ICLR 2027
*A. Bilal, M. A. Mohsin, M. Umer, W. H. Raza, **A. U. Asad**, Y. D. Kwon, M. Valko, D. F. Hougen* [arXiv]

Workshop Papers

[W1] Faithfulness Is Not Free: Auditing Offline KV-Cache Quantization in Retrieval-Augmented Generation GroundLM @ EMNLP 2026
A. U. Asad, A. Bilal*, M. Ali, M. Haseeb, D. F. Hougen* [arXiv]

RESEARCH PROJECTS

Quantized Self-Correction Audit Code
Preregistered study | Python, PyTorch, vLLM, AWQ

- Tested whether an "overthinking" marker penalty fixes the looping that 3-bit AWQ induces in DeepSeek-R1-Distill-Qwen-1.5B on MATH-500 (500 paired traces, five arms); quantization alone raised the loop rate by 0.56 and cut accuracy from 0.85 to 0.39.
- The penalty was statistically indistinguishable from a one-line repetition penalty on loop rate and accuracy, and reaches only 49 of its 50 markers and 42.9% of probed contexts.

Re-derivation Leakage in Multi-Agent Unlearning Code

Preregistered study | Python, PyTorch, open-unlearning, TOFU

- Showed that shared memory enforcing a retrieval-time ID blocklist and deletion over the derivation closure can still store forgotten TOFU facts: a node written from an agent's parametric answer has a fresh ID and no parent edges, so both invariants certify it clean.
- With NPO forget10 agents, realistic routing delegated only 3 of 400 forget-set questions.

Natural Language Processing Code

Python, PyTorch, HuggingFace Transformers, FAISS, NLTK

- Fine-tuned XLM-RoBERTa on Urdu–English code-mixed tweets (+4.2% F1 over mBERT) and BioBERT for NCBI Disease NER; built a DPR and FAISS open-domain QA pipeline on Natural Questions.

Large Language Models Code

Python, PEFT, vLLM, LangChain, Ollama

- Applied 4-bit LoRA to LLaMA-2-7B and analyzed rank-wise change with CKA and linear probes; benchmarked KV-cache eviction and quantization on LLaMA-3 and Mistral-7B for latency, memory, and quality; built an NLI-based hallucination detector for RAG outputs.

Teaching Experience

COMSATS University Islamabad, Undergraduate Teaching Assistant Spring 2024 – Fall 2025

- **Machine Learning** (Fall 2025), **Natural Language Processing** (Spring 2025): Conducted weekly lab sessions and tutored 100+ students.
- **Data Structures & Algorithms** (Fall 2024), **Object-Oriented Programming** (Spring 2024): Conducted lab demonstrations, evaluated course projects, and held office hours.

Industry Experience

King Revolution (Pvt.) Ltd., Machine Learning Engineer January 2026 – March 2026

- Built end-to-end supervised and unsupervised ML pipelines, from feature engineering to deployment.

FFC, Data Analyst Intern June 2024 – August 2024

- Cleaned, preprocessed, and explored (EDA) operational datasets to surface trends for business processes.

Reviewer and Service

Workshop Reviewer GroundLM @ EMNLP 2026, VLM4RWD 2026
Student Leadership Management Lead, IEEE Computer Society, COMSATS University Islamabad

Technical Skills

Languages Python, C++, SQL, JavaScript, LaTeX
ML and LLMs PyTorch, HuggingFace Transformers, PEFT, vLLM, TensorFlow, scikit-learn, FAISS
Systems Git, Docker, Linux, FastAPI, Django, PostgreSQL, MongoDB

Honors and Achievements

Harvard CS50 Puzzle Day 2026 Solved all 10 puzzles (certificate).
MIT Informatics Program Participant (certificate).
Competitions Meta Hacker Cup, AMD Developer Hackathon, UC Berkeley Competition.

Certifications

Machine Learning Specialization (DeepLearning.AI and Stanford Online), Deep Learning Specialization (DeepLearning.AI), Google Python Programming Certificate, Data Analysis with Python (UPenn).